

EXP028: Parameterized Non-Separable Quantum Walks on Tori

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Abstract

This experiment extends the fixed non-separable coin results of EXP027 (Grover, DFT, Hadamard 4D) to a parameterized $SU(4)$ coin framework on 2×2 and 4×4 tori.

On the 2×2 torus, revivals occur only at $\phi_R = 0$ and 2π , shifting the fundamental period from 4 to 8 at $\phi_R = \pi/2$.

On the 4×4 torus, $\phi_R = \pi/2$ shifts the fundamental period from 12 to 24. Two-phase cancellation ($\phi_R = \phi_U = \pi/2$) restores the original period of 12.

Biased coins (Right/Left only) revive on both tori at multiples of 4.

1 Introduction

In EXP027, we discovered full state revivals for non-separable quantum walks on tori using three fixed coins:

Table 1: EXP027 Fixed Coin Results Summary

Coin	2×2 Torus	4×4 Torus
Grover	4, 8, 12, 16, 20, 24, 28	12, 24
DFT	16	None
4D Hadamard	8, 16, 24	None

2 Grover Coin

The Grover coin is:

$$G = \frac{1}{2} \begin{pmatrix} -1 & 1 & 1 & 1 \\ 1 & -1 & 1 & 1 \\ 1 & 1 & -1 & 1 \\ 1 & 1 & 1 & -1 \end{pmatrix}$$

3 Results: 2×2 Torus

3.1 Base Grover Coin (No Phases)

Fundamental period: $N = 4$

3.2 Single Phase $\phi_R = \pi/2$

Fundamental period shifts to $N = 8$

Table 2: Phase sensitivity on 2×2 torus

ϕ_R	Period N	Status
0	4	Revival
$0.11\pi - 1.89\pi$	—	No revival
2π	4	Revival

3.3 Two Phases: $\phi_R = \pi/2, \phi_U = \pi/2$

Cancellation restores period $N = 4$

4 Results: 4×4 Torus

4.1 Base Grover Coin (No Phases)

Fundamental period: $N = 12$

4.2 Single Phase $\phi_R = \pi/2$

Fundamental period shifts to $N = 24$

Table 3: Phase sensitivity on 4×4 torus

ϕ_R	Period N	Status
0	12	Revival
$\pi/2$	24	Revival
π	12	Revival
$3\pi/2$	24	Revival
2π	12	Revival
Other	—	No revival

4.3 Two Phases on 4×4 : $\phi_R = \pi/2, \phi_U = \pi/2$

Cancellation restores period $N = 12$

5 Comparison

Table 4: Comparison of phase effects

Property	2×2 Torus	4×4 Torus
Base Grover period	4	12
$\phi_R = \pi/2$ period	8	24
Phase sensitivity	Extreme (only 0, 2π)	Moderate (0, $\pi/2$, π , $3\pi/2$, 2π)
Two-phase cancellation	Yes	Yes

6 Conclusion

Key findings:

1. 2×2 torus: $\phi_R = \pi/2$ shifts period from 4 to 8
2. 4×4 torus: $\phi_R = \pi/2$ shifts period from 12 to 24
3. Two-phase cancellation works on both tori
4. Phase sensitivity is extreme on 2×2 , moderate on 4×4

Code Availability

https://github.com/ianfromsligo/EXP028_parameterised_coin_tori_non_separable